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Detection algorithm of regional peak motion based on acceleration sensor

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ABSTRACT

In this paper, an algorithm of regional peak motion detection based on acceleration sensor is designed. Firstly, the acceleration model is processed by triaxial integration method, and then the signal is filtered by Gaussian filter. Finally, the step result is obtained by combining regional peak detection, threshold limit and other methods. Through the analysis of the data collected by the acceleration sensor, we can extract the parameters representing the motion attitude of the object from the single-chip microcomputer system, and dynamically modify the parameter value set by the system according to the current system motion attitude, and compare the value with the current motion attitude parameter, so as to judge whether the current count is effective. At the same time, this device can be well applied in the industrial system to achieve convenient and effective counting function. The results show that the step counting algorithm in this paper has high accuracy.

1. Introduction

With the development of science and technology and the improvement of people's living standards, people pay more and more attention to convenience and intelligent information transmission [1,2]. Intelligent wearable devices are more and more common in people's lives, such as smart watches, smart glasses, smart clothes, etc. Human behavior recognition is a very important direction in intelligent research, which has a wide application prospect [3–5]. According to the classification of human motion information collection methods, the existing information collection technology can be divided into two categories: video motion state recognition technology and acceleration sensor based recognition technology [6,7]. In contrast, using acceleration sensor to recognize human motion has many advantages, such as convenient to carry, not affected by the external environment, low cost, more free data acquisition and so on. Clothing is a necessity of people's daily life. Wearable technology and Internet of things technology are applied to clothing [8]. The research on intelligent clothing which integrates perception, processing and transmission has become a hot spot and trend of market development. Acceleration sensor is a kind of sensor used to detect the acceleration of an object. It can detect the direction of movement of an object and the acceleration in all directions, and convert it into corresponding digital signals. After reading these digital signals, it can realize the recognition of the motion track after the algorithm processing.

Micro electromechanical system is a kind of equipment with very small size, and its internal structure is generally micron or nanometer level. Among them, micro electromechanical acceleration sensor has also become the standard hardware of smart phones, smart watches and various intelligent information products [9]. The iPhone launched in 2007 was equipped with acceleration sensors, which triggered a wave of intelligent device revolution in the world at that time. Through gravity and direction sensing, it has also become a new way of human–computer interaction [10,11]. At present, more and more products which use acceleration sensor and artificial intelligence technology in the market also bring value to people in various fields. In the field of biological health and motion, the application of acceleration sensor has become more extensive. People carry smart phones equipped with acceleration sensors and some wearable products, such as smart bracelets, smart watches and so on. Therefore, the acceleration sensor has been used in human behavior recognition, gesture recognition and other aspects.

With the development of science and technology and the improvement of people's living standards, people pay more and more attention to convenience and intelligent information transmission. Intelligent wearable devices are more and more common in people's lives, such as smart watches. Human behavior recognition is a very important direction in intelligent research, which has a wide application prospect. Acceleration sensor is a kind of sensor used to detect the acceleration of an object. It can detect the moving direction of an object and the acceleration in each direction, and convert it into a corresponding digital signal [12]. After reading these digital signals and processing by algorithm, it can realize the recognition of motion track. The acceleration sensor collects the three-axis acceleration data [13]. It is

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suggested in the literature that the axis with the largest acceleration change should be taken as the effective axis to represent the motion state of human body, and then the data of the effective axis should be used to analyze and calculate the steps. However, because the effective axis of the acceleration will constantly change in the process of motion, the steps are not accurate. Based on the fusion of the three-axis data of the acceleration sensor, this paper proposes a regional peak detection method to calculate the effective steps. This algorithm can effectively improve the accuracy of the step detection of the wrist pedometer, provide more detailed and accurate data for the upper computer software in calculating the user's energy consumption, and provide a better experience for the user.

2. Principle of motion track detection system

2.1. Working principle of acceleration sensor

Different from the general target detection of static image, motion detection algorithm usually uses the context between successive frames of sensors to locate the region of interest. It is often used in tasks such as personnel counting, action recognition, anomaly detection, fall detection, parking detection, target positioning and tracking, etc. The acceleration sensor first started from the pendulum integral gyro acceleration sensor and gemstone bearing pendulum acceleration sensor in the 1940s, and continued to develop in the 1960s and 1970s, producing piezoelectric acceleration sensor, flexible acceleration sensor, electromagnetic acceleration sensor, etc. In the 1980s, the University of California Berkley first used micromachining technology to produce micro silicon single-dimensional accelerometer on silicon chip. At present, typical Microelectro Mechanical Systems (MEMS) technology acceleration sensors include piezoresistive acceleration sensor, piezoelectric acceleration sensor, capacitive acceleration sensor, tunnel current acceleration sensor [14–16]. Any of the above acceleration sensors include the following three parts:

(1) Mass: when accelerated, it produces an inertial force;

(2) Elastic member: it uses force to do some work;

(3) Detector: work done by sensitive force.

The United States is the world leader in MEMS accelerometer processing and application technology. The integrated accelerometer of analog devices, Inc. is a landmark product of micro machining. Compared with non MEMS accelerometer, the cost is greatly reduced, the volume is greatly reduced, and the power consumption is also smaller. There are three main indicators of the sensor:

Technical index of sensitivity: for an instrument, the higher the sensitivity, the better. The more sensitive the output voltage is, the larger the change is, the easier the measurement is, and the higher the measurement accuracy is.

Technical indicators of bandwidth: bandwidth refers to the effective frequency band that can be measured by the sensor. The measurement of vibration requires a bandwidth of hundreds of Hertz. In order to obtain more accurate acceleration data, the larger the bandwidth, the better.

Technical indicators of measurement range: different applications actually produce different accelerations, such as the acceleration generated by running is generally less than 0.3 g, while the acceleration of strategic missile will reach more than 10 g. Different products need to select the appropriate range according to the actual application [17].

MEMS accelerometers are developing towards higher precision and integration, which can meet the application requirements of strategic missiles. At present, the general resolution of acceleration sensor in the consumer market has reached 1×10^{-4} g, and the accuracy is still improving rapidly. The potential accuracy can be improved by two orders of magnitude. High precision, anti-harsh environment and multi-sensor integration of MEMS acceleration sensor is the future development direction [18,19].

The principle of acceleration measurement is based on the second law of Newtonian mechanics: the acceleration of an object is directly proportional to the force it receives, inversely proportional to the mass of the object, and the direction of acceleration is the same as the direction of the combined external force. Eq. (1) is the calculation method of acceleration:

$$a = \frac{F}{m} \tag{1}$$

The accelerometer is generally composed of the inspection mass m and spring in formula (1). The detection quality is fixed and can only move along the input axis. When the carrier Accelerates along the input axis, according to Newton's law, inertia force F = am will be generated to keep the motion state of the detected mass unchanged. Under the action of inertia force F, there will be relative movement between the detected mass and the carrier, which will deform the spring. When the force generated by the deformation of the spring is equal to the inertia force F in the opposite direction, the detected mass and the carrier remain relatively still again, at this time, the deformation of the spring reflects the magnitude of the measured acceleration. The overall flow chart of the system is shown in Fig. 1.

Auxiliary components include support, potentiometer, spring, damper and housing. The potentiometer is connected with the spring. When the spring deforms, the moving contact of the potentiometer displaces and the current of the accelerometer changes, so the acceleration signal is converted into an electrical signal for output. The damper is used to improve the dynamic quality of the accelerometer vibration system, and the support is used to fix the detection mass to move along the sensitive axis, so as to ensure that the acceleration direction is consistent.

Due to the complexity of human daily activities, wrist, big arm, knee joint, ankle and other parts of the movement is more frequent, the data collected from these parts as a feature part is not regular, with a high rate of miscalculation. When the sensor module is placed on the wrist, the wrist will swing back and forth during the walking process, and the acceleration value of each axis will change periodically. In the MEMS acceleration sensor structure, the traditional spring is replaced by silicon cantilever or flexible shaft. In this paper, the sensor uses capacitance detection [20]. When the sensitive capacitor moves, it is pulled by the spring to shift, so the capacitance changes and current output is generated. The electric signal enters the preamplifier circuit, and the signal-to-noise ratio of the signal is increased by the signal conditioning circuit, and then analog-to-digital conversion is performed to obtain a digital signal output [21].

Triaxial accelerometer is a device that can measure the mutually orthogonal acceleration of X axis, Y axis and Z axis at the same time. It can be divided into two types in structure. One is to realize the acceleration of three sensitive axes on the same silicon chip. The simplest way is to make three independent sensitive element structures on the same substrate. The other is to place three single degree of freedom acceleration sensors in an orthogonal way and then package them together. The ADXL345 sensor uses the first method, and Fig. 2 is the basic schematic diagram of ADXL345.

2.2. Motion detection algorithm

(1) Triaxial integration processing of acceleration signal

When the acceleration sensor is worn on the wrist, with the wrist swinging back and forth during walking, the acceleration values of each axis will have obvious periodic changes. In this paper, the acceleration data is modeled to improve the accuracy of the step. The calculation formula is as follows:

$$SVM = \sqrt{V_x^2 + V_y^2 + V_z^2} \tag{2}$$

In formula (1), SVM represents acceleration of triaxial integration, V_x, V_y and V_z represent acceleration data of triaxial integration. As

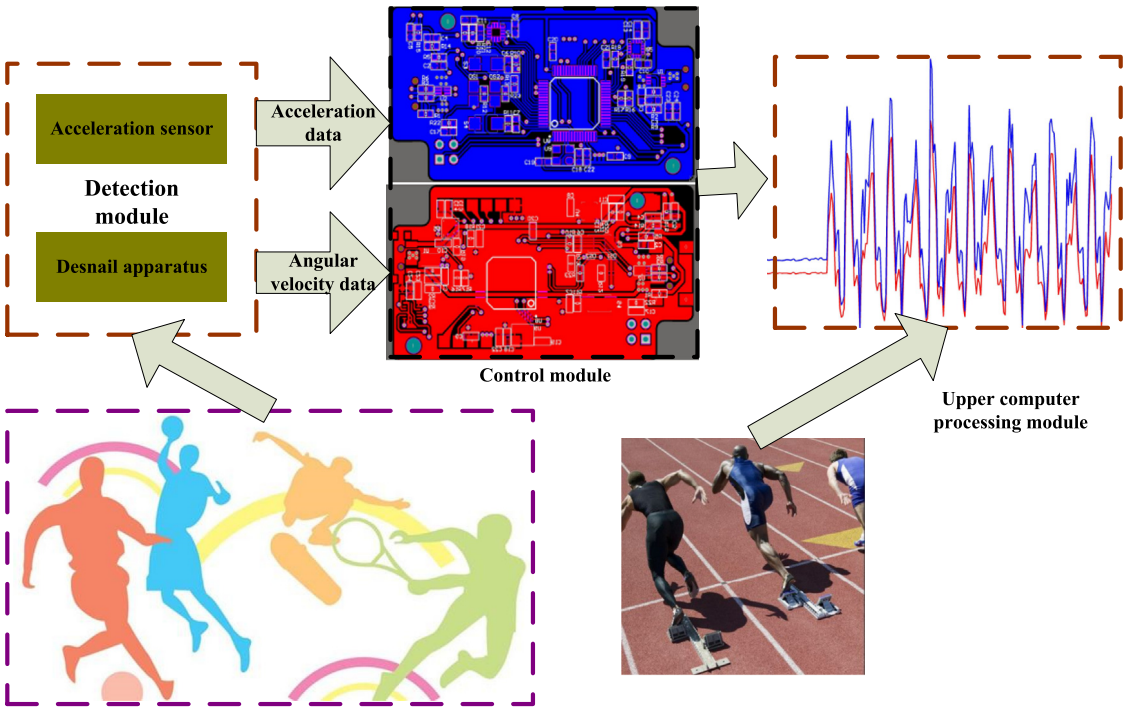


Fig. 1. Overall flow chart of the system.

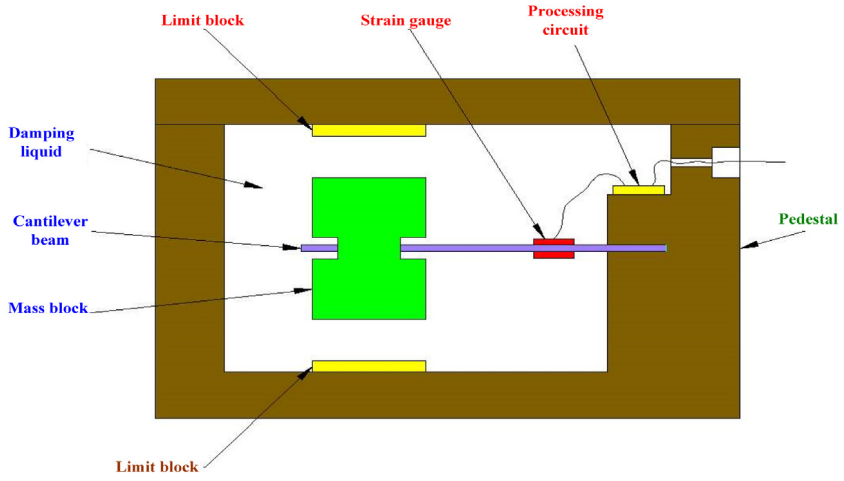


Fig. 2. Schematic diagram of acceleration sensor.

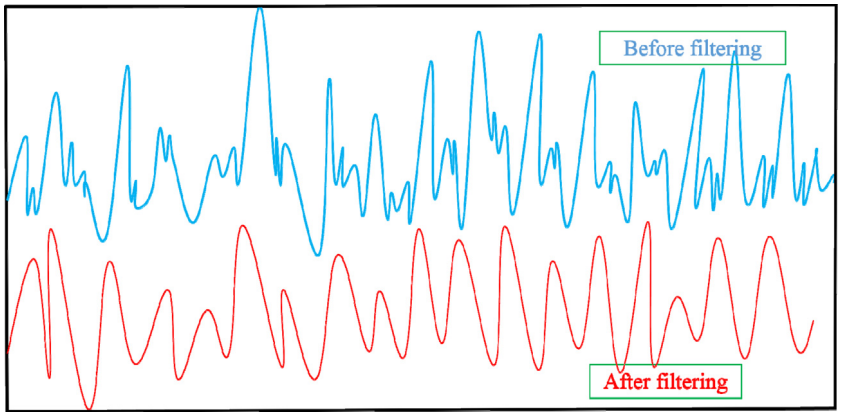


Fig. 3. Comparison of acceleration sensor detection waveforms before and after filtering.

shown in the blue line of Fig. 3, it is the acceleration waveform after triaxial integration processing in walking state.

(2) Filtering processing of acceleration signal

The acceleration signal is filtered to remove noise with smooth waveform. In this paper, the Gauss filter is used to filter, and the one-dimensional Gauss filter generates the weight of the points in the specified domain through the following one-dimensional zero mean Gauss function, that is, to generate the Gauss template. Gaussian filtering is shown in formula (3):

$$a_{filted}(t) = \sum_{t=1}^{S-1} e^{-\frac{t^2}{2\sigma^2}} a(t) \tag{3}$$

Among them, $a(t)$ is the acceleration signal filtered by the above low-pass filter, and a_{filted} is the acceleration signal filtered by Gaussian filter. The parameter σ mainly determines the smoothness of the waveform. The larger σ is, the wider the frequency band of the Gaussian filter is, and the better the smoothness is. Gauss filter is a weighted average with weight. It uses Gauss distribution to generate weight, which makes the weight of the center larger than that of the adjacent points, and can better retain the original wave peak and valley after filtering.

In this paper, the Gauss template size is 50 points, and the standard deviation σ size is 10. The acceleration waveform filtered by Gauss is shown in the red line of Fig. 3.

(3) Wave crest detection

When people are exercising, the fastest step is not less than 0.2 s, and the slowest step is not more than 2 s. Therefore, when calculating steps, the steps whose time interval between the two steps is not within 0.2–2 s are judged as invalid steps, which are not included in the steps of walking.

The difference between the peaks and troughs of the adjacent acceleration waves can be used as the judgment basis to exclude some invalid peak and valley data 0. The calculation formula is as follows:

$$CSVM = a_p(t) - a_v(t) \tag{4}$$

Among them, $a_p(t)$ and $a_v(t)$ represent the acceleration of the adjacent wave crest and wave trough respectively. When the difference between peak and valley value is greater than 1.96 m/s^2, it can be regarded as an effective walking data, which can be included in the step statistics, otherwise, it is regarded as invalid jitter.

3. Detection of motion attitude

3.1. Representation of motion attitude parameters

The detection of motion attitude is based on the result of data analysis after getting the sensor. In this process, how to express a motion feature with parameters is the most important. In human senses, the motion of objects is expressed in the form of continuous images, which is not realistic for single-chip microcomputer. For the application of single chip microcomputer, which has relatively few resources, we can extract the attitude features of motion and make fuzzy matching according to each feature value.

In this device, the following parameters are used to represent the motion attitude:

(1) Moving spindle

The main axis parameter of the motion indicates which direction the current object is moving in. Because of the acceleration sensor, it can determine which direction the object is moving according to the acceleration of the three axes. Based on this, we can judge whether the actual movement conforms to the main axis direction of the movement, as shown in Fig. 4.

(2) Maximum acceleration

The maximum acceleration can represent the intensity of the object's motion. When the acceleration is positive, the longer the time is, the greater the speed is. Although the acceleration decreases, the speed still increases. When the acceleration is zero, the speed reaches the maximum. Through the acceleration parameter, we can limit the extent of the object's motion. In order to effectively express the validity of the data, we usually calculate the maximum acceleration during the two zero points of the line, as shown in Fig. 5.

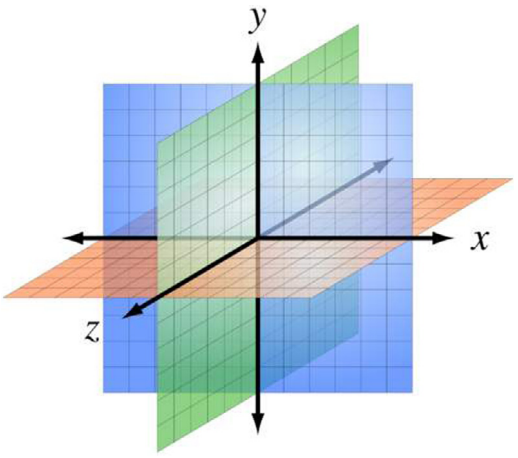


Fig. 4. Main axis of motion.

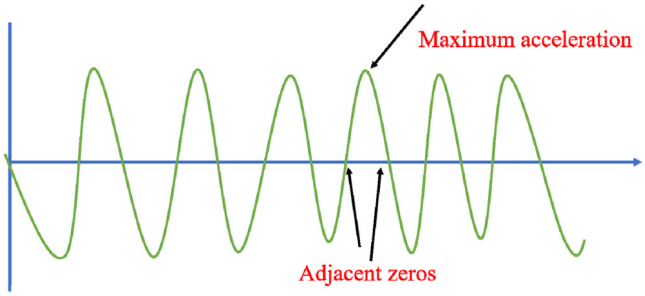


Fig. 5. Maximum acceleration.

(3) Time interval of zero crossing

The time interval of zero crossing represents the continuous time interval between two adjacent zero points, which is used to filter the jitter of some data and ensure that the left data can truly express the actual movement, as shown in Fig. 6.

3.2. Motion attitude recognition

The recognition of motion attitude is mainly based on the fuzzy matching of motion attitude parameters, so as to realize the intelligent recognition of motion trajectory and count.

The acceleration of the current system in each axis is acquired by sensor data, and the time between each axis at zero point is calculated, and the time interval from the last time to zero point is calculated. At the same time, calculate the maximum acceleration in this time interval, and take the direction of the maximum acceleration of the three axes as the main axis of this movement, and compare with the main axis direction set by the system, and then judge whether the movement direction of the object conforms to the set value. If the set value is met, compare the time interval of taking the adjacent zero point with the maximum acceleration and the preset value. If both of them are met, then the current sensor's count value is valid, and the flow is shown in Fig. 7.

3.3. Motion posture self-learning

The self-learning of motion attitude is to enable the system to automatically obtain the set parameters. The design idea is to give the

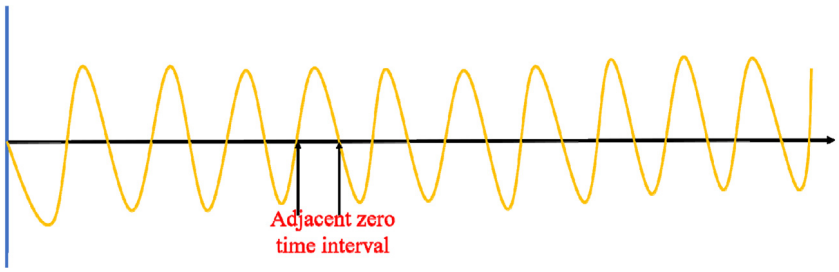


Fig. 6. Time interval.

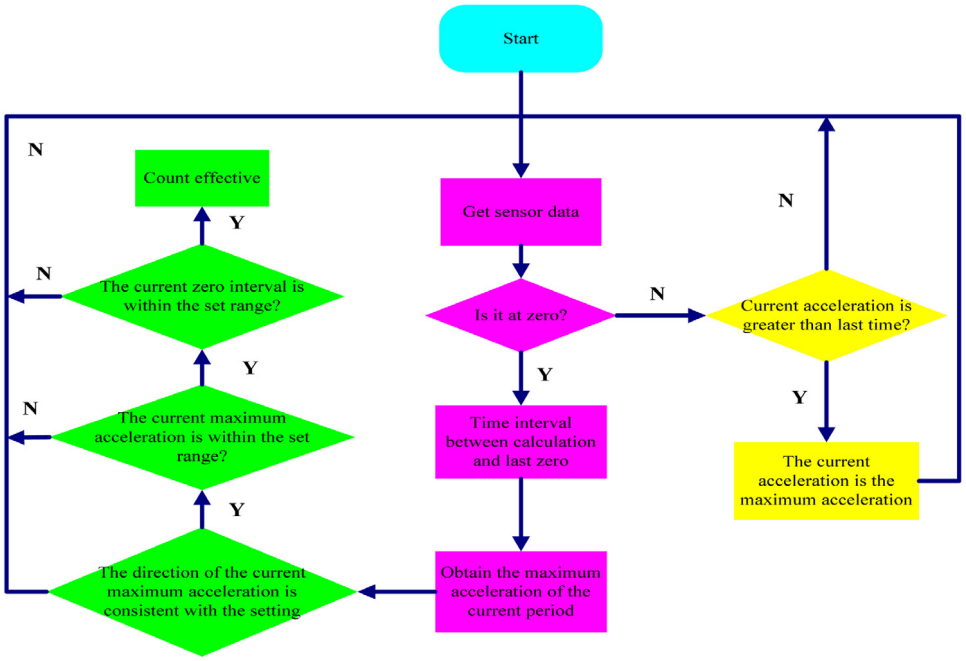


Fig. 7. Attitude recognition flow chart.

system an initial parameter. In the future, all the parameters that the system depends on come from the operation of the system, so that the system can continuously adjust the parameters according to the periodic motion law in the process of operation.

Since the system is basically periodic motion with strong regularity, the motion eigenvalues generated by the system in operation are almost the same. The system can modify the original parameters according to the parameters generated periodically. The principle of modification is to filter the parameters generated in each cycle with IIR. The operating parameters of the system will depend on the results of each operation of the system. For the results of periodic operation, each operation is to strengthen the results of this operation.

4. Experiment and analysis

4.1. Static state data acquisition and calibration

Collect the acceleration data under static state and eliminate the 0 g offset of the sensor. The Table 1 and Fig. 8 shows the three-axis data output of the sensor under the static state of the horizontal desktop. The room temperature is 21 °C, Vs = 3.3 V. Calculate the mean value of the data and get the 0 g offset correction value.

The three-axis calibration value is stored in the drive SYS attribute calibrate, and the 0g offset can be eliminated every time the sensor data is read.

Table 1
Sensor 0 g offset data.

Acceleration sensor data (g)		
X axis	Y axis	Z axis
−0.0040	−0.0335	1.0263
−0.0029	−0.0345	1.0313
−0.0104	−0.0235	1.0235
−0.0044	−0.0306	1.0235
−0.0079	−0.0395	1.0220
−0.0077	−0.0226	1.0225
−0.0077	−0.0313	1.0328
−0.0029	−0.0306	1.0306
−0.0032	−0.0290	1.0328
−0.0070	−0.0319	1.0290
Average value: −0.00581	Average value: −0.0307	Average value: 1.02743

4.2. Linear motion accuracy test

Data acquisition of linear motion. Make a linear acceleration motion along the X-axis on the horizontal desktop, move 1.0 m, and conduct low-frequency sampling of 100 Hz, a total of 327 groups of data. Select 10 groups of data and compare them with the data after Kalman filtering, as shown in Table 2. The comparison before and after filtering of x-axis and y-axis is shown in Fig. 9, where mark 1 in the figure represents before filtering and mark 2 represents after filtering.

After the acceleration data is filtered, the data obtained by the random sampling method is used for error compensation. The velocity

Table 2
Data comparison before and after Kalman filter.

Acceleration sensor data before filtering (g)			Filtered acceleration sensor data (g)		
X axis	Y axis	Z axis	X axis	Y axis	Z axis
−0.0079	−0.0075	1.0110	−0.0079	−0.0075	1.0110
−0.0113	−0.0040	1.0021	−0.0120	−0.0044	1.0026
0.0703	0.0203	1.0105	0.0713	0.0215	1.0106
−0.0628	−0.0225	1.0064	−0.0636	−0.0242	1.0068
−0.0315	−0.0120	1.0104	−0.0318	−0.0122	1.0103
−0.0159	−0.0074	1.0070	−0.0179	−0.0043	1.0069
0.0360	−0.0117	1.0063	0.0347	−0.0088	1.0064
0.0084	−0.0140	1.0132	0.0093	−0.0110	1.0149
0.1716	−0.0232	1.0060	0.1722	−0.0202	1.0062
0.1126	0.0080	1.0148	0.1124	0.0046	1.0141

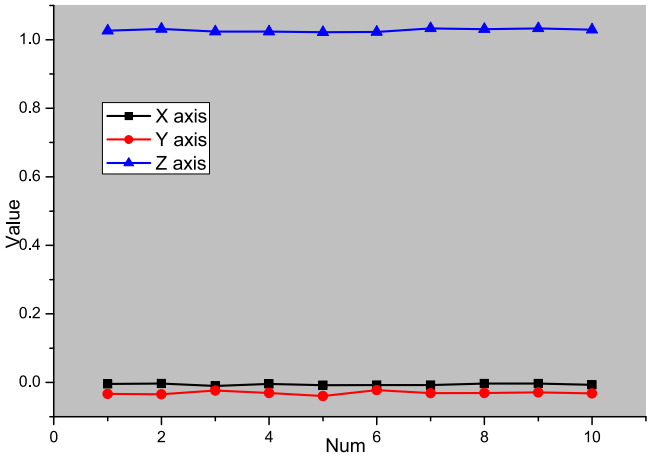


Fig. 8. Sensor offset data.

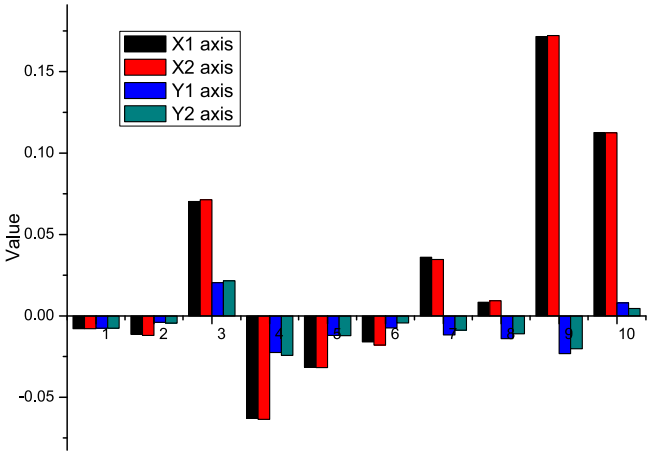


Fig. 9. Comparison of filtering before and after x-axis and y-axis.

and displacement data obtained are the final processing data. Without the error compensation by Kalman filter and random sampling method, the data moves 1.0 m along the X-axis and the measured displacement is 0.8 m.

4.3. Frequency domain analysis

The following uses the method of Fourier transform to calculate the frequency spectrum corresponding to the time domain diagram of slow walking, fast walking and running of human body, as shown in Figs. 7–9.

The frequency spectrum corresponding to the time domain graph is obtained by Fourier transform. The frequency of human motion can be qualitatively analyzed theoretically by analyzing which frequency the original data is composed of Fig. 10 is the Fourier spectrum of the sampling data, the abscissa is the frequency, the ordinate is the amplitude of the frequency, and the magnitude reflects the proportion of the frequency. It can be seen that the main frequency is about 1.9 Hz when walking slowly, about 2.2 Hz when walking fast, and about 2.8 Hz when running. This is because the order of stride frequency from small to large is slow walking, fast walking and running.

Combined with the time and frequency domain analysis of human motion, the combined acceleration of walking slowly and fast can be obtained between 0.5 and 1.5. The main frequency of jogging is about 1.9 Hz, while that of fast walking is about 2.2 Hz. The running acceleration changes from 0 to 2.4, and the main running frequency is about 2.8 Hz.

5. Conclusion

The main research of this paper is the algorithm of motion track detection based on three-axis acceleration sensor. With the development of technology, micro sensors with lower cost have been widely used. Through the analysis of the data collected by the acceleration sensor, we can extract the parameters representing the motion attitude of the object from the single-chip microcomputer system, and dynamically modify the parameter value set by the system according to the current system motion attitude, and compare the value with the current motion attitude parameter, so as to judge whether the current count is effective. At the same time, this device can be well applied in the industrial system to achieve convenient and effective counting function. In the long run, it is necessary to replace the traditional external observation method in many fields to install the acceleration sensor on the target object to detect the motion track. However, due to the problem of accuracy, the current acceleration sensor cannot replace the traditional external observation system. Therefore, the main work of this paper is to study the relevant algorithms and try to improve the accuracy of motion trajectory detection based on acceleration sensor.The error sources of the motion track detection system are divided and studied more carefully, and more efficient error processing algorithms are designed for various errors. Through the acceleration data, the software is used to adjust and compensate the influence of the user interface of the sports team equipment. Object motion detection can accurately detect the mobile phone's motion mode through one or more motion sensors, and dynamically adjust the user interface, such as enlarging the touch area of the virtual button, or changing the display position of each element in the user interface.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

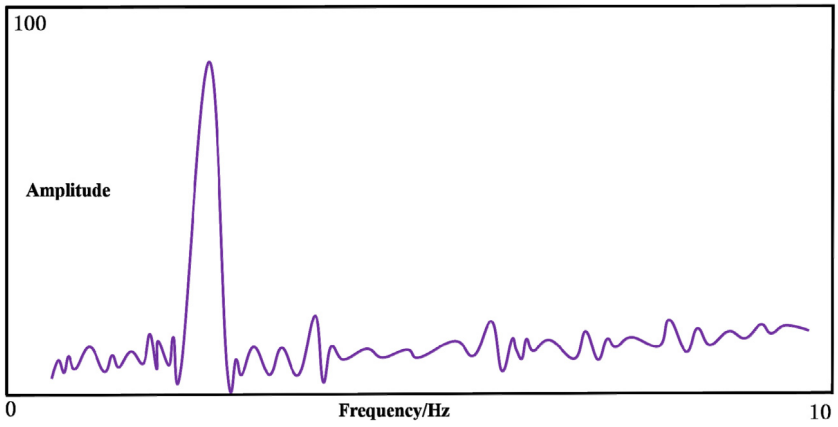


Fig. 10. Spectrum diagram of different motion states.

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